Augmenting Interaction: Collecting Common Sense Through AR Objects

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Abstract. Augmented Reality applications are becoming more popular with the continued miniaturization of technology. With the increasing use of smart phones, which often provide increased processing power, enhanced and open software platforms, Augmented Reality has become instrumental in the way we perceive our surroundings and the information that it carries. It is now possible to implement an Augmented Reality system without carrying bulky and expensive equipment. Currently, there are many systems that implement some form of Augmented Reality to provide a specialized interaction to users. However, those systems usually employ expensive, immobile components with highly specialized interfaces. In this paper we present a novel approach for building interactive interfaces using Augmented Reality. UIAR (User Interface through Augmented Reality) is an augmented reality framework that allows for the ubiquitous creation and dissemination of interactive user interfaces and augmented reality objects. Here we present the novel interaction schemes of UIAR alongside the implementation of the framework's intended purpose - collecting common sense through AR objects.

Keywords: augmented reality, human-computer interaction, common sense, ubiquitous user interfaces

1 Introduction

Augmented Reality (AR) is a fairly young area of research which is currently expanding in many of the already existing fields of Human-Computer Interaction and Computer Interfaces. In our research we implement an Augmented Reality system which will serve as an extension to existing computer interfaces, provide enhanced user-experience, and define virtual objects and their actions in an ubiquitous way.

To implement such a system, however, we must first address three major areas where we think Augmented Reality user interfaces can be improved.

Firstly, Augmented Reality objects are hard coded into applications, which makes them highly specialized and not ubiquitous. Secondly, there is no standard for defining Augmented Reality interfaces and how they react to human interaction. Finally, there is no implementation of natural interaction with such interfaces and objects.

In the field of collecting common sense knowledge data from a large number of voluntary users, one of the biggest challenges is to keep the users engaged, entertained, and focused so as to collect a sufficiently large amount of high-quality data. Most such projects start with a somewhat big user-base, which eventually dwindles in numbers and activity as the project grows older. We believe using UIAR to implement games through its interactive framework can prove beneficial in this area.

We begin by presenting a summary of the related research, connecting our approach with previous work. Next we present use cases which describe specific functionalities that our framework enables and describe the purpose of the framework. This is followed by a description of the software tools and libraries we take advantage of while implementing the framework. Next we present the proof-of-concept games we have implemented with UIAR to collect common sense. We conclude by giving short review of the steps ahead.

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2 Related Research

The existing research into Augmented Reality and Human Computer Interaction that is relevant to this study can be divided in three areas: finger and hand-based interfaces, paper-based interfaces, and Augmented Reality applications. We will describe each one briefly, followed by a review of approaches to collecting common sense.

2.1 Hand and Finger-based Interfaces

The technologies for hand and finger-based interfaces can be roughly split in two categories - sensing-based and computer-vision based. Sensing-based systems like [21] are very robust but are often limited to detecting only "touch" behavior, not able to recognize hands or other physical object that come into view. Computer-vision based systems like [5], [6], [17] are often limited by the lighting conditions and may not respond well to sudden changes in the field of view. However, systems like [10], [11], [8] have proven to be robust and accurate enough. We are using a computer-vision based system since wearing special hardware to enable "touch" capability reduces the mobility of the system.

2.2 Paper-based Interfaces

Most of the existing paper-based interfaces fall into three categories - using paper alongside digitizing tablets like [16], using digital paper technologies like [4], and using paper tagged with markers (barcodes, fiducial markers, etc.) like [17]. In our system we will be using only 2D paper tags (AR markers) for 3D positioning of the visual objects, unlike [11] where the hand position and direction is used to determine the position of the virtual object. The interaction between the user and the interface will be entirely virtual or conveyed through AR marker motion.

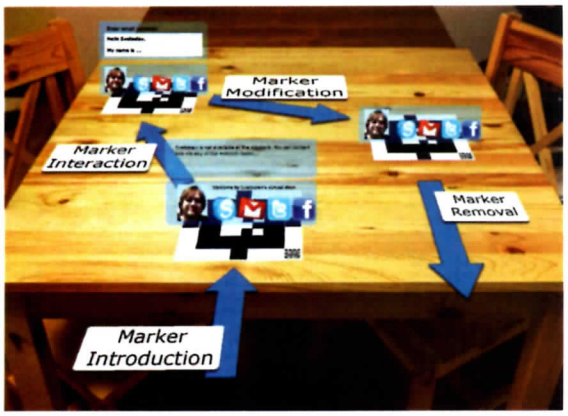


Figure 1. Example usage scenarios

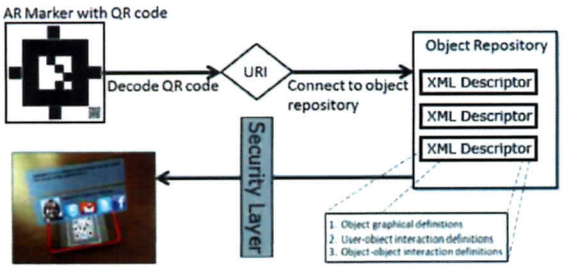


Figure 2. Basic framework model

2.3 Augmented Reality Applications

There have been many Augmented Reality applications, using either multiple-camera hand and object tracking or a single camera (like a webcam). Those applications vary in both their mobility and complexity. Our project was inspired for the most part by the Sixth Sense project developed by Pranav Mistry in the MIT Media Lab [18]. As is the purpose of [18], we strive to provide mobility, affordability and ubiquity to Augmented Reality applications.

There are two major differences between the paradigm employed by [18] and our framework, which highlight the novelty of UIAR. We let the user utilize any device to view the AR environment (mobile phone, webcam plus desktop, AR goggles, etc.) where [18] projects the AR environment over objects themselves. We also use 2D paper AR markers to determine correct 3D coordinates and scale for object placement.

The system presented in [20] showcases a collaborative AR environment, allowing multiple participants to interact with two and three-dimensional data using tangible user interfaces.

2.4 Augmented Reality Frameworks and Authoring Tools

With the popularization of software libraries like ARToolKit [7] there has been a lot of development to bring AR authoring tools in the hands of researchers and developers. However, frameworks like DWARF [2] and osgART [15] are quite complex and require an expert programmer. Our framework on the other hand gives developers with enough programming experience in ActionScript3 the ability to construct and distribute user interfaces, interactive objects, etc. with ease.

2.5 Common sense acquisition

One of the great challenges facing the artificial intelligence community is creating agents that can operate and adapt in a natural human environment. Naturally, we must be able to provide those agents with the information and the learning tools necessary for them to operate. If we want them to be able to make decisions about, relate to and have a simple understanding of the global environment in which they function, they need to be provided with basic knowledge about the world [12]. In humans, this knowledge is available to us in the form of reasoning shortcuts and factual information about each particular occasion we find ourselves in, and is generally known as common sense.

This type of knowledge is also crucial for fooling humans that they are conversing with another human - in other words passing the Turing test.

With the realization of the importance of common sense to the field of artificial intelligence, considerable research has been done towards collecting and structuring this type of knowledge. The biggest research effort by far has been the Cyc project [12] which has already collected over a million common sense assertions in little over two decades. As the project became more of a commercial venture, a much smaller set of data is available free of charge. The work required, however, has been considerable. Common sense knowledge is manually input by experts in particular areas, who first give a complete ontological structure to the data, using a specially developed knowledge representation language called CycL, and then insert domain specific data based on their expertise [12].

Another attempt to collect common sense data is the Open Mind Common Sense project. OMCS collects common sense statements from untrained volunteers over the Web in the form of natural language statements [23]. In the course of few years the project had already collected over 1.6 million statements.

Other systems, like Verbosity [1] and Common consensus [14] identified and addressed one of the major problems with such systems - user interest. In order to consistently gather quality knowledge from a large set of volunteers, they must be given enough motivation to continue to participate, especially in light of the fact that the number of volunteers drops over the life of the project.

Yen-Ling Kuo et al. [9] have successfully utilized social games to collect common sense and their findings provide useful suggestions for designing community-based games.

3 Usage Scenarios

The usage-scenarios described below serve to describe specific features of our framework that are not available in current AR systems. Via those usage scenarios we want to illustrate the particular unique functionalities that our framework provides.

3.1 Universal Marker Registry

In our system we use 2D paper markers for virtual object placement. Currently, there are multiple ways to create such a marker with the only restriction that the pattern not be too complex. This improves

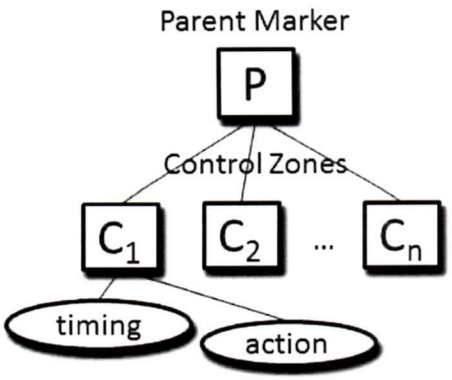


Figure 3. AR Object Control Scheme

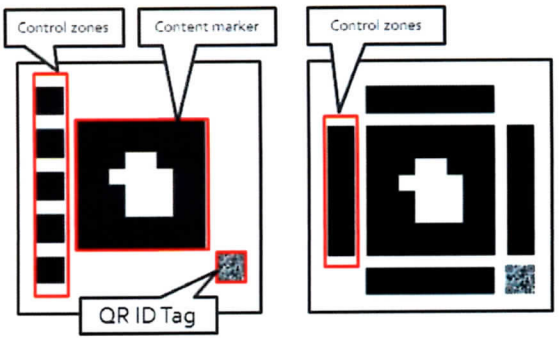


Figure 4. User-object interaction

the marker recognition which in turn allows for a scalable AR experience. The relation between the virtual object and the marker pattern is embedded in the software.

We plan to implement a universal AR marker registry so that information about the object is stored globally. Using this registry, users will be able to point their AR device to any AR marker and display its contents, regardless of whether they have seen the marker before or not. The system will recognize the URL encoded in the marker, download and display the virtual object. We are planning to include a QR code within the AR marker pattern and extract the URL from it.

This will allow for more ubiquitous AR applications. The user will no longer be restricted to using markers specifically designed for his AR system. We hope the implementation of such registry will attract interest from the Augmented Reality community and help construct a large ecology of AR objects. It will also enable developers to construct their own AR Marker ecologies, independent of the main system.

3.2 AR Object-object and Object-user Interaction Definition

The next step in our system is defining virtual object actions as part of their registry information described in the previous section. This way the AR system will know both what objects to display, as well as how those objects are supposed to interact with the user and other objects.

For the most part our virtual objects are user-interfaces. As such, their actions are defined either as user initiated or object initiated. We would like to implement both ubiquitous object-object and user-object interaction. Object-object interactions will allow us to define how virtual objects behave when in proximity of one another. A simple example would be two virtual objects positioned by 2D markers on the field of view, both objects representing a single Skype chat window with different users. If both markers are positioned close to one another the resulting action will be to open a single Skype window making a conference call to both users. User-object interactions will be described as the services the virtual object can perform upon user actions. For example, a virtual object displaying information on a person (a virtual business card) can provide information upon request, provide an email interface, a Facebook or Twitter message interface, current location, etc.

Building such a repository of objects will provide both a functional and a graphical description of the AR objects in an Augmented Reality environment which in turn will make AR applications more ubiquitous.

4 Framework purpose and novelty

- **Ubiquity of Object presence**
  Objects associated with one AR marker must remain the same independent of system used to view the AR environment. Objects are first registered via a QR code which encodes the URI of the repository from which the object graphics model and other data is to be drawn. This gives each AR marker (independent of its graphical representation) an unique identifier. This unique identifier allows us to implement the next three objectives.
- **Object Persistence**
  Objects must carry associated data and object states across different AR environment viewers. Object data and object states are stored in the database defined via the AR marker's URI.
- **Ubiquity of Object Interactivity**
  Objects must behave the same way independent of AR environment viewer. Object's interaction definitions are stored in the database defined via the AR marker's URI.
- **Definition of Object Interaction Models**
  Interaction models with AR systems have so far been system/application dependent. Each system defines for itself how users interact with the AR objects and the interaction model cannot be extended or redefined.
  Our framework allows developers to define how a specific AR Object will interact with the users and with other AR Objects introduced to the scene. They do so by assigning behaviors to extra control markers associated with the AR Object via the AR Object's URI. Figure X shows an example of AR Objects and controls.

Figure 1 shows an example usage scenario encompassing all four focus areas described above. As the AR marker is inserted into the scene, the marker's graphical and functional representation is obtained from the object repository on the server encoded in the QR code. In this case the AR marker's object is a simple business card showing a photograph, 4 buttons which when activated would present

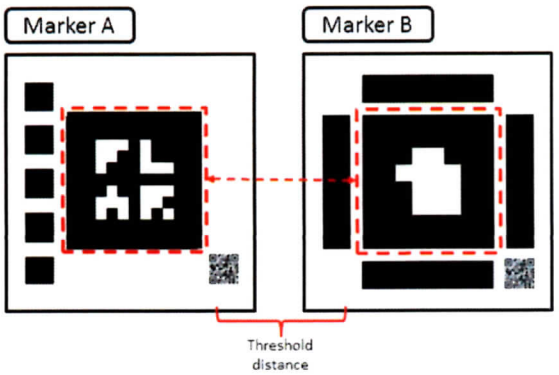


Figure 5. Object-object interaction

a different interface depending on the button, and a message board. The user can modify the content - in this case leave a new message on the message board or modify the object the marker represents - in this case by closing the message board section. Once the marker is removed from the scene it's object properties are saved and the next time the marker is introduced it will remember them. The currently existing implementations of augmented reality systems, although providing some of the above mentioned features, do not encompass all of the features UIAR intends to provide. This, in our opinion, is what distinguishes our framework from the existing ones, and gives UIAR an innovative perspective on augmented reality interfaces.

5 System design, software components and implementation

5.1 System setup

Currently, UIAR is intended to be used as a desktop application. The system has three major components - a desktop PC, a webcamera capable of minimum 640x480 resolution and printed AR markers. The test setup involves positioning the webcamera above and behind the user, who is sitting by a flat surface, with the desktop positioned in front of the user. An alternative setup is to have the webcamera positioned over and perpendicular to the flat surface. As image processing is prone to errors in different lighting conditions, we found it best to have the setup lit up by one uniform light source, away from direct sunlight. This way marker tracking and recognition is optimal. The user can then proceed to introduce markers to the scene, register them through their QR codes and interact with them.

5.2 Software components

Our system is based on several existing technologies that allow us to perform AR overlay, QR decoding, marker recognition, tracking and handling and draw our interfaces programatically. In this section we will look at each one in more detail.

- **AR overlay:** The original AR toolkit was first developed by Dr. Hirokazu Kato from the University of Washington [7] and is currently supported by the Human Interface Technology Lab at the University of Canterbury in New Zealand [3]. As we are building our framework in Adobe ActionScript programming language, we are using a language port of the ARToolKit to AS3 provided by Saqoosha [22], Nyatla [19] and Sparklib [13] named FLARToolKit.
- **QR decoding:** For decoding QR codes in ActionScript we use the QR library provided by Sparklib [13].
- **Marker handling:** To manage marker registration efficiently for multiple markers and predict marker motion we use the FLARManager 0.7 toolkit which is provided by Eric Socolofsky [24].
- **Interface Design:** To design, draw and define our interfaces we use PaperVision3D library provided by [25]. PaperVision3D is a set of libraries that give ActionScript developers a 3D engine for Flash.

All of the above mentioned libraries are distributed under licenses allowing developers to use them free of charge for non-commercial purposes. Our framework is built using Flash Builder 4 and ActionScript 3.5 SDK. Developers can produce their modules using any ActionScript compiler as long as they run the same SDK and use the same versions for PaperVision3D, FLARToolKit, and FLARManager.

5.3 System model

Figure 2 describes the basic model of our system. Here we will look at each component.

- **AR Marker with QR:** We designed our AR markers to include QR codes encoding the Unique Resource Identifiers for the object that the AR marker identifies. This allows the developer to define his own AR marker patterns and objects independent of the viewer. It also allows the AR environment viewer to recognize AR markers without the need to include the patterns in the program. The QR code can be placed either inside of the AR marker as part of the pattern or on the back of the AR marker. Note that if the QR code becomes a part of the AR marker's pattern it must do so in an asymmetrical fashion, since AR marker patterns must be asymmetrical to enable correct marker detection.
- **Database:** The database component of the system implements a simple MySQL scheme with database entries containing developer information and pointing to a local directory for specific marker id. The physical file is a precompiled Adobe SWF file that contains the AR Object's graphical and interaction definitions.

With object persistence we ensure that an AR object will retain its information and state in case it is removed from the AR environment. As we saw in Figure 1 , if an AR Marker is introduced and the user makes a modification to the state of the object it represents, the system will relate that change to the database. The next time that marker is introduced to the scene, the system will display it's previously modified state.

5.4 System interaction

In order to continue to the next section we must define the control scheme for AR Objects. Figures 3 and 4 show how we implement interaction with our objects. In the database, each parent AR Marker has associated with it a set of control patterns that define a single action. The control patterns are additional shapes printed on the side of the AR marker. Those control patterns are our equivalents of a "button". Each control pattern is defined by a "timing" parameter and an "action" parameter. The system continuously scans for the patterns

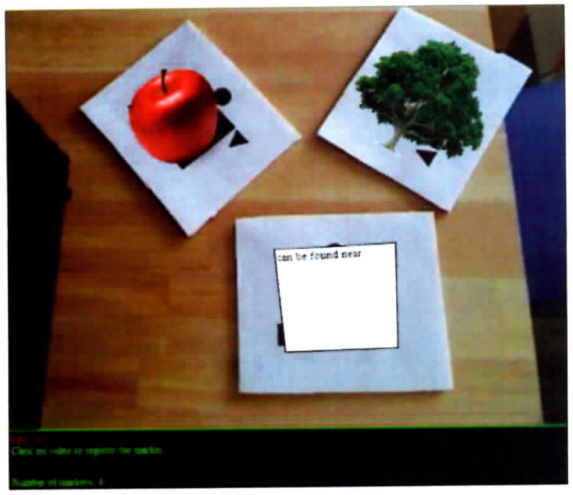


Figure 6. Visual Sentence Patterns

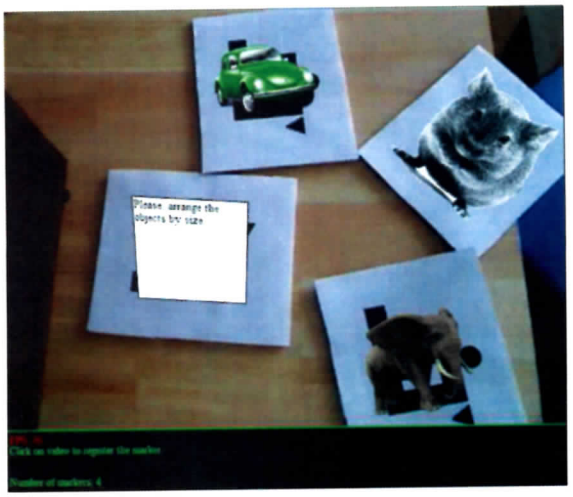


Figure 7. Arrange by Feature

and detects if the control has not been visible longer than the "timing" parameter specifies (which means that the control is activated), and performs the action based on the "action" parameter.

We can now describe how our system's user-object and object-object interaction paradigms.

5.4.1 User-object Interaction

Figure 4 describes how the users will interact with the AR Objects. For each marker ecology (defined by the database the system is connecting to) there will be multiple control patterns active. The shapes representing the control patterns do not have to be unique, since they are registered by the developer upon uploading the marker into the UIAR database. The detection technique the system uses is the same described in section **5.4**.

5.4.2 Object-Object Interaction

The last type of interaction we define in our system is object-object interaction. Figure 5 gives an example of one AR object being aware of another. In this paradigm the developers of the AR Objects are allowed to define a relative "distance threshold", which serves as a trigger for a predefined action. Since at any given time the system knows which patterns are present on the scene, along with their relative size, it can calculate relative distance between each marker. A simple example of this interaction scheme is illustrated in the next section, where aligning all markers in close distance simulates the action of a "submit" button.

6 Common sense acquisition games

We believe that using our framework can prove useful when it comes to both acquiring new volunteers and keeping the existing volunteers interested and engaged in the process. Using the UIAR framework we can implement games that are rewarding, engaging, and interactive. Moreover, those games can be targeted towards younger audiences who naturally spend more time playing games.

In this section we present a proof-of-concept implementation of UIAR for the purpose of collecting common sense.

With UIAR, developers can assign any virtual object to an AR marker (3D models, textual and media content, etc.) which will be immersive (objects will blend in with the actual environment) and interactive (objects will be aware of the surrounding objects and react to users' input). Here we present 3 implementations of games to collect specific types of common sense inspired by such games as Verbosity [1] and Common Consensus [14]. Images for those games are acquired through Google Image Search. We use Wordnet to choose related concepts when needed.

6.1 Visual sentence pattern game

Figure 6 gives an example of an instance of the visual sentence pattern game. One type of exercise commonly used to collect common sense is to simply fill in the gaps in a sentence pattern. While natural language sentences can often prove difficult to process, the use of different sentence patterns allows for collecting data that can be disambiguated, categorized and easily parsed.

Our example game uses templates such as: "X is a kind of Y", "X is used for Y", "X is typically found near/in/on Y", " X is the opposite of Y", "X is related to Y", etc. A user is provided with AR markers, which after being registered via their QR codes will correspond to X, Y, and a description of the template respectively. As the visual content of AR markers is dependent on the QR code only, the game can be setup so that every time the markers representing X or Y is re-registered, the content is changed.

For example, on first registry marker X can hold a 3D model of an apple and Y a 3D model of a tree. If the sentence pattern does not make sense, the user can re-register it until he gets the correct one (in this case, "X is typically found near/in/on Y"). To submit the entry, the user needs to arrange all three markers so that they touch each other. After submission, the game refreshes the markers with new objects and sentence patterns and the user can keep playing.



Figure 8. Arrange in Sequence

6.2 Arrange by feature game

In figure 7 we can see an example of the second type of game which is oriented toward spatially oriented knowledge collection. In this case the number of markers/objects can be as many as the screen can allow. A sample exercise of this game would be to ask the user to arrange the markers in a certain order based on a certain criteria (height, length, size, etc.). As the markers are spatially aware of each other, the user will complete the exercise by putting the markers close to each other in the order needed.

For example, the user can have 4 markers. After registering each marker, he is presented with 4 different 3D (or 2D) objects which he/she will arrange by a predefined feature and submit to the system. The same game can be used to cluster individual objects in case there is more than one feature. For example, the user can be presented with representatives of fruits and animals, in which case he can group the markers together and choose their categorization.

6.3 Arrange in sequence game

The third type of game, illustrated in Figure 8, is goal oriented. Just like the "arrange by feature" game, in this game the user will be presented with two markers representing the beginning and the end of an activity, with the rest of the markers representing actions that must fit in a sequence.

For example, the user can start with 2 markers, one showing - the rising sun (or a person coming out of bed) and another a steaming cup of coffee. The rest of the markers could represent "boiling water", "mixing water in cup", "opening coffee", "pouring sugar". The user will complete the exercise by arranging the markers/activities in the right order.

Each object/activity can be represented either by a visual (a 2D or 3D model) or just text (the text being overlaid over the AR marker). In order to represent both simple physical objects and abstract concepts with more visual appeal it is better to use images or models. The example implementations given are specific to the realm of common sense acquisition. However, the system can be generalized to serve any number of language acquisition tasks.

7 Conclusions

In our research we are trying to address the need for enriching textual knowledge with interaction driven knowledge acquisition. We plan to implement an AR viewer for mobile devices using an HTC developer device running Android 2.3 OS. Additionally, we plan to implement additional methods for user-object interaction, improve the overall usability of the system, and implement security schemes for the AR objects.

We are currently in the process of deploying a prototype version of the UIAR framework as an open source project. We are performing evaluations on the common sense knowledge collection games, the quality of the collected data and how our system affects user retention.

We view Augmented Reality as a technological path that will keep extending and developing with constant future software and hardware improvements. We believe Augmented Reality, as opposed to Virtual Reality and Augmented Virtuality, is the medium that will be most appealing to everyday users. The medium through which we collect, interact and distribute knowledge, will diverge from the nowadays common desktop/laptop/smart-phone solution. Data interaction will become more ubiquitous as the devices through which we perform everyday tasks become themselves more ubiquitous. Even today there are multitude of projects and proof-of-concept products that look into the future of how we interact with machines. They are only one step away from becoming as mass produced and spread as the smart phone has become in the last 5 years. This proliferation of ubiquitous devices, not restricted by common keyboard/display solutions are the reason why research into interactive AR interfaces is so important to us.

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